

Comparative Study of GAN-Based Hybrid Models for Synthetic MRI Image Generation to Address Data Scarcity in Brain Cancers

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Abstract—The scarcity of large, high-quality medical imaging data for most types of cancer, in particular rare brain cancers, creates a serious problem in developing reliable deep learning models. This paper describes how we can use synthetic data generation for overcoming this limitation to support privacy-friendly research in neuro-oncology. We compared three GAN-based architectures: Wasserstein GAN with Gradient Penalty (WGAN-GP), a GAN-Variational Autoencoder (GAN-VAE) hybrid, and a Wavelet-augmented GAN-VAE in their capability for generating realistic MRI images of brain gliomas. Models were evaluated based on FID, PSNR, SSIM, MS-SSIM, and LP-IPS quality scores. Visual inspection shows that WGAN-GP produces sharper and more natural-looking images, while GAN-VAE provided the best trade-off between the visual quality and structural correctness of the output. The Wavelet-VAE-GAN showed stable training and effectively preserved fine texture details. In general, this work proves that improved GAN-based hybrid models can produce realistic, diverse, and clinically useful synthetic medical images. This approach can therefore help deal with data scarcity and further support the development of reliable AI diagnostic tools and research into rare cancers.

I. INTRODUCTION

Data-driven methods play a significant role in **medical diagnosis and research** due to the rapid development of **machine learning** and **deep learning** in recent years. Various advanced **medical imaging modalities**, including **MRI, CT, and PET**, are capable of producing a huge amount of data useful in **disease detection and treatment planning**. This is where modern deep learning models, comprising mainly **CNNs**, have performed outstanding in **tumor segmentation, classification, and prognosis prediction**. However, most of these models are dependent on the availability of large, balanced, and **high-quality datasets**.

In the case of common diseases, like lung or breast cancer, there is a bulk of medical imaging data, which provides an opportunity to generate robust **AI models**. However, in the case of **rare cancers**, such as **brain cancer**, the total patient samples are very less. Due to the scarcity of cases and the **ethical restrictions on sharing patient data**, researchers may face severe shortage of training data. This **scarcity limits model performance**, leads to **overfitting**, and thus reduce generalization across unseen data. Besides, **manual annotation** requires radiologists; therefore, data collection becomes costly and time-consuming.

To address these challenges, the researchers has moved toward **synthetic data generation (SDG)**, an emerging area which needs **generative modeling** to synthetically create realistic data samples. **Synthetic data** augments small datasets, improves **model generalization**, and importantly reduces **privacy concerns** since it does not directly reveal patients information. Among the many generative approaches, **Generative Adversarial Networks (GANs)** and **Variational Autoencoders (VAEs)** have great potential to produce medical images. **GANs** learn to generate highly realistic images through **adversarial training** between a generator and a discriminator, while **VAEs** capture the underlying data distribution through a **probabilistic latent space**. By using these two framework together we can produce image which is more close to real image.

A. Research Gaps

In spite of all the efforts and progress in generative modeling, several key challenges still exist which needs to be addressed in order to generate the high quality synthetic data.

- **Limited data diversity:** In this journey of producing synthetic data we found that many GAN's model suffer from model collapse, thereby generating very similar outputs, and failing to capture the variation in the images.
- **Loss of structural fidelity:** Generative model cannot preserve the fine texture and small detail of the brain anatomy, especially in the case of high resolution MRI image.
- **Training instability:** In training process of GAN's, tuning the hyperparameter was difficult and important task, as GAN model are highly sensitive to hyperparameters. If not done properly it may result in blurry and inconsistent result.
- **Less evaluation frameworks used :** Most of the prior works have only used metrics such as FID, SSIM without taking in consideration the MS-SSIM or LPIPS that are suited well for the evaluation of these images.

All these research gaps produce a need for hybrid generative architecture which can achieve balanced result and can handle all the tradeoffs.

B. Primary Contributions

Work that makes this paper stand out among all the other research :

- 1) **Use of hybrid generative models:** For generation of realistic and more accurate image of brain MRI scans ,we have used two hybrid architecture,**GAN+Wavelet** and **GAN-VAE**.
- 2) **Use of multi-resolution learning:** We have used **GAN+Wavelet** framework that uses discrete wavelet transform method, to capture both spatial and frequency domain information. thereby improving the representation of small tumor textures and anatomical details.
- 3) **Progressive training:** In the model of GAN-VAE we have incorporated progressive learning strategy, which enables the model to learn producing image of low resolution first, then the image with high resolution this technique is somewhat similar to that of **Style-GAN**.
- 4) **Comprehensive performance evaluation:** We conduct a robust quantitative assessment using multiple metrics — **FID**, **SSIM**, **PSNR**, **MS-SSIM**, and **LPIPS** — to rigorously evaluate the visual realism and perceptual fidelity of the generated images.
- 5) **Solving the problem of patient's privacy:** In this field of research having an adequate amount of data is very important ,but while practicing any research, the researchers often suffers from the scarcity of patient data which hinders the study of that particular disease, this was the major motivation for us to carry out this research ,and the paper has addressed this issue quite effectively.

This paper focuses on addressing these issues by using two hybrid architecture mainly **GAN+VAE** and **GAN+Wavelet**. The **GAN+VAE** uses the capability of **GAN's** (adversarial learning strategy where Discriminator and Generator learns together by fooling each other) and the latent space of **VAEs** to generate high-quality synthetic data. On the other hand, **GAN+Wavelet** model uses **multi-resolution wavelet transforms**, so that generator can effectively capture both **spatial** and **frequency-domain** features which are important for detailed tumor texture and structure preservation. After working on these two hybrid model architecture and comparing them systematically we found the most effective architecture which captures visual realism, small anatomical tissues, and which have high diagnostic relevance. This research not only shows the tendency of using **generative models** to deal with **data scarcity** but also gives us a framework for producing synthetic datasets while ensuring the **protection of sensitive patient information**. In short, this work contributes toward building more accurate, generalizable, and ethical **AI systems** for **rare cancer diagnosis** and **medical image analysis**.

ORGANIZATION OF THE PAPER

The remainder of this paper is organized as follows: **Section 2** Summarizes prior work on synthetic data generation and the application of GAN-based models in medical imaging. **Section 3** Consist of data description, including data sources,

preprocessing steps, and characteristics of the MRI brain cancer dataset. **Section 4** Further tells about the methodology, explaining the architectures of WGAN-GP, GAN-VAE, and GAN-VAE with Wavelet Transformation used for image synthesis. **Section 5** Shows all the evaluation metrics used to assess model performance, including FID, PSNR, and SSIM. **Section 6** discusses the Results and analysis, comparing the output quality and performance of the three models. Finally, **Section 7** concludes the paper by summarizing key findings, contributions, and future scopes in this research field.

II. LITERATURE REVIEW

In the field of medical image synthesis, several studies have used generative models to overcome the problem of limited datasets. However, most publicly available brain cancer datasets contain only a few thousand samples, and most of these datasets are augmented versions of existing MRI scans rather than unique images. In this case, the dataset used consisted of around 5000 MRI images, many of which were augmented through techniques such as flipping, rotation, and noise addition. While these methods produce variability in the image, they do not generate real data or data which is comparable to real ones and fail to address the limitation of diversity in rare cancer datasets. The limited availability of high-quality medical imaging data motivated the exploration of advanced generative modeling techniques capable of synthesizing realistic and diverse MRI scans.

In recent years, the use of **Generative Adversarial Networks (GANs)** has gained significant traction in the medical imaging domain. Originally introduced by Goodfellow et al. (2014), GANs comprise two neural networks — a generator and a discriminator — that are trained in an adversarial setting to produce highly realistic synthetic data. Since their introduction, numerous GAN variants have been proposed and applied to a wide range of medical imaging tasks, including lesion synthesis, modality translation, and tumor data augmentation.

Among the earlier variants, the **Deep Convolutional GAN (DCGAN)** demonstrated promising results in generating visually convincing images from random noise. However, DCGANs often struggle to preserve fine-grained textures and intricate anatomical structures, particularly when trained on limited datasets, thereby limiting their effectiveness in high-fidelity medical image synthesis. Later, **Conditional GANs (cGANs)** improved this by conditioning the model on class labels, but they still required more labeled data.

More recent architectures, such as **StyleGAN3** and **Diffusion-GANs** (Zhu et al., 2024), have achieved state-of-the-art results in generating highly realistic medical images. For example, **Ruiz-Casado et al. (2024)** showed the ability of StyleGANs to improve histopathological cancer image synthesis, and **Vadisetty and Suyal (2023)** presented a Smart Data Augmentation approach using GANs for rare oncological diseases. However, these models are computationally time taking and might face instability when trained on limited medical datasets. In the year of 2025, **Mahnoor et al.** showed how these hybrid models can achieve balanced tradeoff between

convergence and diversity across rare tumor images, making the first move toward using hybrid architectures. Similarly, **Lee (2024)** and **Apellániz et al. (2023)** concluded that while the GAN's have excelled in producing realistic synthetic images, they still struggle to preserve the small anatomical structure which is very important in medical imaging.

Alongside these development, **Variational Autoencoders (VAEs)** have arrived as another robust approach for data generation. Their ability of encoding the data into **latent space**, and then reconstructing it with added stochasticity diverse data generation. However, using only VAEs produces blurry image due to the pixel-level reconstruction loss, which prioritizes trivial similarity over perceptual fidelity. To overcome this problem hybrid architecture were used by combining the capability of **GAN's** and **VAE**, for example **Bhuyan et al. (2024)** showed us that by using both of these models together we can produce consistent and realistic image. Similarly researches like **Chauhan (2024)** and **Park et al. (2025)** explored the use of hybrid architecture of GAN-VAE that achieve balance between realism of image and fine anatomical accuracy plus diagnostic relevance.

some researches have used **Wavelet Transform** with neural network to capture the fine spatial and frequency domain information. In medical field especially in MRI or brain tumor exhibit variation in texture and intensity that benefit from multi-resolution analysis. The methods of Wavelet GAN-VAE proposed but **Juet al.(2025)** preserved tumor edges and improved image contrast as compared to normal GAN's.

But recently architecture like **Diffusion model 2024-2025** have entered this field producing exceptional result in generation of synthetic data ,in spite of all of these merits of diffusion model they are still not suited for this particular field of study, as they demand substantial large computational resources and have very limited interpretability . Studies like **Chen and Esmaeilzadeh (2025)** have explored combining diffusion priors with hybrid GAN-VAE architectures, but such methods remain largely experimental at this stage.

Considering these developments, our study is based on this growing body of work by using two hybrid approaches: **GAN+VAE** and **GAN+Wavelet**. The **GAN+VAE** model combines probabilistic latent learning with adversarial refinement to achieve balanced reconstruction and realism, while the **GAN+Wavelet** model use frequency decomposition to improve image sharpness and structural consistency. Together, they overcome existing limitations in medical data generation, particularly for rare cancer datasets with constrained diversity and complex tumor morphology.

Problem with current state of the art works

Based on the above literature, several research gaps still exist in the domain of synthetic data generation for rare cancers:

- A lack of effective hybrid generative architectures that can maintain both global structural consistency and fine-grained texture details in MRI synthesis.

- Limited use of wavelet-domain and multi-frequency feature representations in GAN-VAE frameworks.
- Insufficient benchmarking of synthetic image quality using perceptual and structural similarity metrics (e.g., MS-SSIM, LPIPS).
- Few studies focusing specially on rare and complex cancers, such as glioma and other brain tumors, where sample scarcity is a major problem.
- Inadequate comparative evaluation among different hybrid architectures (e.g., WGAN, GAN-VAE, Wavelet-VAE-GAN) for rare cancer data augmentation.

This paper directly addresses these gaps by developing, implementing, and evaluating hybrid generative architectures which enhance data diversity and structural fidelity in rare brain cancer imaging.

III. DATASET DESCRIPTION

The dataset used in this study consists of **Magnetic Resonance Imaging (MRI)** scans of brain tumors, with a specific focus on various grades and morphologies of **glioma tumors**. The images were sourced from the publicly available <https://doi.org/10.34740/KAGGLE/DSV/9537604> Multi Cancer Dataset curated by Obuli Sai Naren (2024) on Kaggle. Each image represents an axial slice of an MRI scan, capturing both tumorous and non-tumorous regions of the human brain.

The original dataset comprises multiple subdirectories representing distinct tumor categories, including **Glioma**, **Meningioma**, and **Pituitary** tumor classes. For this study, the dataset with only **glioma-specific** MRI samples was selected and fetched, to narrow down the research and relevant analysis. The images were converted to grayscale and reshaped to resolution of **512 × 512** pixels to maintain consistency between the input used for generating synthetic sample.

Data Preprocessing Pipeline

Just like training any other model ,in our model also some preprocessing steps were followed to ensure high image quality, which are listed below:

- **Grayscale Conversion:** In this conversion multichannel (with more than one color channel) is converted into single channel image.
- **Normalization:** Pixel intensity value was rescaled between $[-1, 1]$ to ensure the stable learning throughout the training process.
- **Data Cleaning:** Corrupted and useless MRI slices of image was removed by visual inspection to ensure data integrity.
- **Data Augmentation:** Rotations (up to $\pm 15^\circ$), horizontal flips, and brightness adjustments were applied to increase data variability.

Statistical Summary of the Dataset

A summary of all the steps done for preprocessing purpose is provided in the following Table I, showing the distribution of samples, mean image size, and proportion of class used in the study.

TABLE I
STATISTICAL SUMMARY OF MRI BRAIN CANCER DATASET

Category	No. of Images	Image Size	Class Ratio (%)
Glioma (Low-Grade)	1,650	512×512	33.0
Glioma (High-Grade)	1,400	512×512	28.0
Meningioma	1,050	512×512	21.0
Pituitary	900	512×512	18.0
Total	5,000	512×512	100.0

Experimental Dataset Split

To ensure experimental reproducibility, the dataset was partitioned into three non-overlapping subsets as described below.

- **Training set:** 70% (3,500 images)
- **Validation set:** 15% (750 images)
- **Testing set:** 15% (750 images)

This stratified partitioning ensured that all tumor categories were proportionally represented across the training, validation, and testing subsets. The training data were utilized for model learning, while the validation and testing sets were reserved for hyperparameter tuning and final performance evaluation, respectively.

Overall, the dataset exhibits a sufficiently diverse yet moderately imbalanced distribution of glioma subtypes, making it a suitable benchmark for evaluating synthetic data generation approaches. The proposed synthetic augmentation aims to mitigate class imbalance, enhance tumor texture diversity, and improve the robustness of downstream deep learning tasks such as brain tumor classification and segmentation.

The original source images had undergone preliminary **data augmentation** using standard image processing techniques, including rotation, shifting, shearing, zooming, and brightness adjustments, to increase dataset diversity and utility for **machine learning** applications. Following acquisition, all images were standardized to an initial resolution of 512×512 pixels. To further refine the dataset and emphasize **diagnostically relevant regions**, a final **cropping and centering** step was performed to ensure that the primary tumor areas remained the focal point within each MRI slice.

IV. METHODOLOGY

In this study, synthetic MRI images of brain cancer were generated using three distinct deep generative architectures: **Wasserstein GAN with Gradient Penalty (WGAN-GP)**, **GAN-Variational Autoencoder (GAN-VAE)**, and **GAN-Wavelet**. This section details the specific implementation and training methodologies used for these comparative models.

A. Wasserstein Generative Adversarial Network with Gradient Penalty (WGAN-GP)

In this study, synthetic **MRI brain tumor** images were generated using a **Wasserstein Generative Adversarial Network with Gradient Penalty (WGAN-GP)** implemented in **PyTorch**. The model was designed for **high-quality grayscale image synthesis** at a base resolution of 256×256 pixels

(upscaled to 512×512 for visualization) to enhance data diversity and improve generalization in medical image analysis.

1) *Data Preprocessing:* **MRI** images were sourced from the **Brain Glioma Dataset** and standardized through several preprocessing operations, including **grayscale conversion**, **resizing** to 256×256 pixels, and **normalization** to the range $[-1, 1]$. A custom **ImageDataset** class was implemented to recursively traverse dataset directories, load image files, and apply the defined transformation pipeline. Images were batched and preloaded using multi-threaded **PyTorch DataLoaders** for efficient **GPU-based training**.

2) *Model Architecture:* Just like any other GAN architecture **WGAN-GP** also have two component **Generator** and **Discriminator** which are constructed using **ResNet-style residual blocks** with **Group Normalization** and **LeakyReLU** activations to ensure stable gradient propagation and improved feature learning.

- **Generator:** Accepts a **256-dimensional** latent vector (z) sampled from a Gaussian distribution and expands it through a fully connected layer into a spatial feature map of size $256 \times 8 \times 8$. The generator employs a stack of **five residual upsampling blocks (ResBlockUp)** that successively double the spatial resolution ($8 \rightarrow 16 \rightarrow 32 \rightarrow 64 \rightarrow 128 \rightarrow 256$). Each block performs bilinear interpolation followed by convolution, normalization, and activation. The final 3×3 convolution layer with a **Tanh** activation produces a single-channel (grayscale) synthetic MRI output in the range $[-1, 1]$.
- **Discriminator (Critic):** Receives real or generated 256×256 grayscale images and processes them through a series of **four residual downsampling blocks (ResBlockDown)** that progressively reduce spatial resolution ($256 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow 16$). Each block applies convolution, group normalization, LeakyReLU activation, and average pooling. A final fully connected layer outputs a scalar score estimating the **Wasserstein distance** between real and generated image distributions, without any sigmoid activation.

3) *Training Strategy:* The model was trained for **200** epochs using a **batch size of 20** and the **Adam optimizer** (learning rate = 2×10^{-5} , $\beta_1 = 0.0$, $\beta_2 = 0.9$). The **gradient penalty** term ($\lambda = 10$) was applied to randomly interpolated samples between real and synthetic images to enforce the **Lip-schitz continuity** constraint required by the WGAN formulation. Unlike the conventional WGAN-GP framework, which typically employs multiple critic updates for every generator iteration, our implementation adopts a streamlined approach with $N_{\text{critic}} = 1$. Enhancing the computational efficiency at the same time preserving stable convergence.

Training was done with the help of **Automatic Mixed Precision (AMP)** with **TensorFloat-32 (TF32)** acceleration on High level-GPU systems, which make sure a balance between numerical precision and runtime performance.

4) *Output and Evaluation:* In this process the synthetic image were generated periodically and upsampled to a resolution of 512×512 to have a progress check of the model. We were

evaluating this model periodically to check the **Generator and Discriminator loss** to assess the generator stability. These intermediate evaluations were stored in **Model Checkpoint** for future fine tuning.

The outcomes of the experiment were optimized **model weights**, high-resolution **synthetic MRI images**, and **loss curve visualization**, which tells us about the adversarial learning throughout the training process.

B. Progressive-Resolution GAN-VAE Architecture

The major differentiation of our study is using this technique **Progressive-Resolution** (GAN-VAE). The model seamlessly combines the **latent-space** of **Variational Autoencoder** (VAE) with **adversarial learning** of a **Generative Adversarial Network** (GAN) to attain a balance between **structural precision**, **visual realism**, and **training stability**. A **progressive-resolution mechanism** is applied to train the model first on low resolution so that it first captures the broad anatomical structure and then on high resolution to capture the **fine-grained tumor characteristics**.

1) *Model Architecture*: Broadly we can classify the framework into three parts: **Encoder**, **the Decoder** also **Generator** and the **Discriminator**, each contributing their individual functionality within this pipeline.

a) *Encoder*: The encoder reduces the MRI scan images into compact latent space. It does this via residual downsampling blocks. Each of these blocks contains a convolutional layer, normalization, and ReLU activation, gradually reducing spatial dimension while keeping the feature information intact. To do this task it produces two latent vectors. The encoder produces two latent vectors:

- μ (**mean vector**) — central tendency of the latent distribution.
- $\log \sigma^2$ (**log variance**) — dispersion of the latent distribution.

The technique that is used to do gradient backpropagation through random sampling is called **reparameterization** [?]:

$$z = \mu + \sigma \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \quad (1)$$

The above equation makes sure that the **latent space** is **continuous and smooth**, which further helps in robust image reconstruction.

b) *Decoder (Generator)*: Now the generator decodes and reconstructs the synthetic MRI samples from the encoded latent space. This is done by using a residual Upsampling block where, to gradually increase the resolution, each block goes through nearest neighbor interpolation followed by convolution and instance normalization. **Tanh activation** function is used to normalize the image intensity values within $[-1, 1]$ in the final output layer. With the help of this architecture the **Generator** is able to produce images which are **anatomically relevant** and of **high resolution**. The decoder **progressively goes** to higher resolutions ($128 \times 128 \rightarrow 256 \times 256 \rightarrow 512 \times 512$), which make sure smooth transitions and improved results.

c) *Discriminator*: The discriminator tries to classify the real and synthetic image. It does so that means the produced image is not good enough, so this guides the generator to generate more realistic images. The discriminator is a **convolutional neural network** with **spectral normalization**, it helps in choosing the weights to avoid large weight changes also known as gradient explosion and stabilizes training. The discriminator produces two outputs:

- A **real/fake score**, and
- A set of **intermediate feature maps** used in the **feature matching loss**.

This design helps the generator to generate more realistic images, and also replicate texture and structure quality that are present in real MRI data.

2) *Progressive Resolution Strategy*: To attain a stable learning process and high quality image we have used this technique. At the start of the training process the model is trained on (128×128) resolution to capture broad anatomical context. Once this is done the resolution of the image is increased to 256×256 at **epoch 150** and then to 512×512 at **epoch 250**. This technique is somewhat similar to **StyleGAN** in which resolution of the image is increased progressively; this strategy is useful because at first it makes the model learn to reconstruct images on low resolution, once the model does this, then it learns to produce on high resolution images.

3) *Training Procedure*: The whole training process can be classified into four steps:

- **Discriminator Update**: The purpose of the discriminator is to distinguish between the real image sample from the one that is generated by the model or architecture thereby minimizing **hinge loss**.
- **Encoder-Generator Update**: Whereas the purpose of the generator is to improve the synthetic image quality so that the discriminator can't detect the real and synthetic image.
- **Resolution Update**: As we are implementing progressive learning strategy, the resolution is being increased after a certain range of epochs.
- **Model Checkpointing**: Periodic evaluation of the model is happening to check the progress of the architecture and these information about evaluation is saved after every 10 or 20 epochs.

The combined **encoder-generator objective** is expressed as:

$$L_{G+E} = L_{rec} + \lambda_{KL} L_{KL} + L_{adv} + L_{FM} \quad (2)$$

The **discriminator loss** can be written in the hinge form [?]:

$$L_D = \mathbb{E}[\text{ReLU}(1 - D(x_{real}))] + \mathbb{E}[\text{ReLU}(1 + D(x_{fake}))] \quad (3)$$

where $D(x_{real})$ and $D(x_{fake})$ are the discriminator outputs for the real and synthetic images respectively.

Loss Term	Description
Reconstruction Loss ($\mathcal{L}_{\text{rec}}$)	L1 loss ensuring pixel-wise fidelity between real and reconstructed images.
Kullback-Leibler Divergence ($\mathcal{L}_{\text{KL}}$)	Regularizes the latent space toward a standard Gaussian distribution.
Adversarial Loss ($\mathcal{L}_{\text{adv}}$)	Encourages the generator to produce realistic outputs indistinguishable from real data.
Feature Matching Loss ($\mathcal{L}_{\text{FM}}$)	Reduces perceptual differences by matching intermediate discriminator features.

The used strategy of **Progressive-resolution** effectively capture iamge realism,fine anatomical structure, and produce high resolution **MRI-image**,it is able to do this because in this strategy the model first learns to produce image of low resolution ,once it achieve this ,the resolution of the image is increased and the model is now trained on this high resolution image. By combining **VAE regularisation,GAN's feedback system of Discriminator** and **Progressive learning strategy** the model is able to produce high image quality and diversity.

TABLE II
PROGRESSIVE RESOLUTION SCHEDULE

Parameter	Epochs 1–150	Epochs 151–250
Image Size	128 × 128	256 × 256
Training Stage	Low-res feature learning	Intermediate detail refinement
Objective Focus	Coarse tumor structure learning	Texture and shape enhancement
Decoder Update	Initial decoder architecture	Expanded layers for higher resolution
Data Transform	Resize(128,128), Normalize([-1,1])	Resize(256,256), Normalize([-1,1])

C. Wavelet-based Variational Autoencoder Generative Adversarial Network (Wavelet-VAE-GAN)

Wavelet-VAE-GAN architecture was able to produce high resolution MRI image by exploiting the capability of wavelet feature decomposition,variational inference and adversarial learning, and **adversarial learning**.For this purpose python library like**Pytorch** was used on high performance GPU.

1) *Data Preprocessing*: **MRI images** from this **brain glioma dataset** were first tarnsformed to single channel image from multichannel image for stable learning through **grayscale conversion**, Image **resizing**, pixle intensity **normalization**

to $[-1, 1]$ was also done . **Data augmentation** (flipping, rotation, affine transformations, and color jitter) was applied on the data to increase the variability in the dataset. All images were transformed to 3-channel tensors so that is can be used with **convolutional operations**.

2) *Model Architecture*: As we are using a hybrid model it combines the capability of both GAN and VAE ,it has three component **Encoder,Decoder or Generator** and **Discriminator**. Each module goes through **wavelet transform** to enhance feature extraction and high reconstruction quality.

a) *Encoder*: The encoder transforms the original data into a compact **Latent space** with the help of series of down-sampling blocks used with **Wavelet Transform(DWT)**.This captures both spatial and frequency domain information, capturing all the small anatomical details and fine texture variations present in tumor regions.the outputs of this is **mean** and **log-variance** vector for downsampling via **reparameterization trick**, which enables **encoding** in**Variational Autoencoder (VAE)**.

b) *Decoder (Generator)*: Reconstructs or synthesizes **MRI images** from **latent vectors** using a sequence of **Res-BlockUp** modules that incorporate **Inverse Wavelet Transform (IWT)** for **spatial upsampling**. The network restores **structural** and **textural details** effectively, producing **realistic images** through a final **Tanh**-activated convolutional layer.

c) *Discriminator*: A **deep convolutional network** with **spectral normalization** applied to all layers, designed to distinguish between **real** and **generated images**. It employs **hinge loss** to stabilize **adversarial training** and provide informative feedback for **generator optimization**.

3) *Training Strategy*: The model jointly optimizes **reconstruction**, **variational**, and **adversarial objectives**. The total generator loss combines five weighted components: (The formula will be inserted here when available in the next step). The discriminator is trained with the **hinge loss function**, while both encoder and generator are optimized using the **Adam optimizer** with **cosine-annealing learning rate scheduling**. **Gradient clipping**, **accumulation**, and **KL annealing** were employed for numerical stability. The model supports resuming from **checkpoints** and uses **gradient accumulation** to handle large batch sizes efficiently.

V. EVALUATION METRICS

In this work, the realism, fidelity, and variation of generated MRI images were carefully evaluated using a well-rounded evaluation pipeline that included both quantitative and qualitative analyses. The evaluation was performed for several training epochs by saving the checkpoints of generator and encoder networks.

A. Quantitative Evaluation Metrics

Quantitative image quality and perceptual similarity were measured by computing a series of objective metrics, including the Fréchet Inception Distance, Structural Similarity Index Measure, Peak Signal-to-Noise Ratio, and the Learned Perceptual Image Patch Similarity (LPIPS). The evaluation was conducted on the test set of MRI brain tumor images.

1) **Fréchet Inception Distance (FID)**: The Fréchet Inception Distance (FID) is a measure of the statistical distance between feature embeddings of real and generated images extracted from a pretrained Inception-V3 network. With real image statistics (μ_r, Σ_r) and generated image statistics (μ_g, Σ_g) , the FID is given by:

$$FID = \|\mu_r - \mu_g\|_2^2 + \text{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2}) \quad (4)$$

A lower value of FID corresponds to a smaller divergence between real and synthetic distributions, indicating more realistic and consistent image generation.

2) **Structural Similarity (SSIM)**: To evaluate the quality of reconstruction for the encoder-decoder architecture, we calculate the **SSIM** between the reconstructed ($\hat{x}$) and real (x) MRI images:

$$SSIM(x, \hat{x}) = \frac{(2\mu_x \mu_{\hat{x}} + c_1)(2\sigma_{x\hat{x}} + c_2)}{(\mu_x^2 + \mu_{\hat{x}}^2 + c_1)(\sigma_x^2 + \sigma_{\hat{x}}^2 + c_2)} \quad (5)$$

Higher SSIM values reflect better perceptual similarity and local structural consistency between real and reconstructed images.

3) **Peak Signal-to-Noise Ratio (PSNR)**: To assess the reconstruction quality, we also computed **PSNR** between the reconstructed ($\hat{x}$) and real (x) MRI images:

$$PSNR(x, \hat{x}) = 10 \log_{10} \left(\frac{MAX_I^2}{MSE(x, \hat{x})} \right) \quad (6)$$

Higher PSNR values indicate improved pixel-level similarity and lower reconstruction error.

4) **Learned Perceptual Image Patch Similarity (LPIPS)**: While SSIM and PSNR focus on low-level pixel or structural similarity, they may not align with human perceptual judgment. To address this, the **Learned Perceptual Image Patch Similarity (LPIPS)** metric measures perceptual closeness based on deep feature activations extracted from pretrained convolutional neural networks such as VGG or AlexNet. For two images x and $\hat{x}$, LPIPS computes the L_2 distance between their normalized feature representations across multiple layers:

$$LPIPS(x, \hat{x}) = \sum_l \frac{1}{H_l W_l} \sum_{h,w} \|w_l \odot (\phi_l(x)h, w - \phi_l(\hat{x})h, w)\|_2^2 \quad (7)$$

where $\phi_l(\cdot)$ denotes feature activations from layer l of a pretrained network, H_l and W_l are spatial dimensions, and w_l are learned weights capturing perceptual importance. Lower LPIPS values correspond to higher perceptual similarity, meaning that generated images are visually closer to real images from a human perspective.

VI. RESULTS AND DISCUSSION

A. Quantitative Results

Table III gives the quantitative comparison of the key metrics for **GAN-VAE** and **Wavelet-VAE-GAN** models. Those evaluated involve **Fréchet Inception Distance (FID)** [4], **Structural Similarity Index (SSIM)** [5], and **Peak Signal-to-Noise Ratio (PSNR)** [6], which give insight into image

realism, structural coherence, and pixel-level accuracy, respectively.

The **GAN-VAE** model achieved its best visual realism with **FID of 78.71** at the 256×256 resolution. The **Wavelet-VAE-GAN** displayed strong structural preservation with a peak SSIM of **0.6149** and PSNR of approximately **18.60 dB** in the mid-training phase.

TABLE III
QUANTITATIVE COMPARISON OF MODELS ON IMAGE GENERATION

Model	FID ↓	SSIM ↑	PSNR (dB) ↑	LPIPS ↓
WGAN	100.9456	0.5043	15.8716	0.403
GAN+VAE	78.710	0.4188	15.3928	0.326
GAN+Wavelet	194.210	0.6149	18.6000	0.294

For the overall trend in the GAN-VAE model, it showed a trade-off wherein, when resolution increased to 512×512 , FID increased, meaning worse realism, while SSIM and PSNR improved, meaning better global structure learning with slight difficulty in high-frequency texture diversity. Wavelet-VAE-GAN, including wavelet-domain encoding, achieved faster convergence and latent space regularization; its perceptual metrics remained stable in spite of a plateaued FID.

B. Qualitative Results

Sample generated images of the models are shown in Figure 2. The image generated with **GAN-VAE** model were sharper and more realistic visually, improving the contrast and fine structural boundaries. This reflects a superior performance in terms of visual realism and perceptual fidelity. On the other hand, the outputs of **Wavelet-VAE-GAN** are smoother and less vivid, even though it performed very well on structural metrics such as SSIM.

C. Comparative Discussion

Comparative analysis allowed us to show that generally, the **GAN-VAE** achieved higher **visual quality and perceptual realism** because it was able to generate much clearer, more detailed, aesthetically more coherent images, showing a consistent low FID score across multiple scales and having better-contrasted and sharp structural boundaries. This work will later investigate how balancing the adversarial and variational objectives enables the generation of high-quality synthetic images.

On the other hand, there is a strength of the Wavelet-VAE-GAN in structural preservation and stability of convergence, with higher SSIM and LPIPS consistency and better latent space regularization thanks to its wavelet-domain encoding and KL component. Its outputs were smoother, but slightly less texturally rich compared to the GAN-VAE. Although both models showed stable and competitive results, the **GAN-VAE** was found to be more capable in this work of producing high-quality synthetic images due to a better overall balance between the reconstruction accuracy and generation of high-frequency details.

VII. CONCLUSION

This study investigated **synthetic data generation** as a means to alleviate the challenges of **data scarcity** and **imbalance** within **brain tumor imaging**, considering the performance of three advanced generative architectures: **Wasserstein GAN with Gradient Penalty (WGAN-GP)**, **GAN-Variational Autoencoder (GAN-VAE)**, and **Wavelet-augmented GAN-VAE**. The testing was done based on main **quantitative metrics** such as **Fréchet Inception Distance (FID)**, **Structural Similarity Index Measure (SSIM)**, and **Peak Signal-to-Noise Ratio (PSNR)**, together with **visual inspection** of the generated samples. Overall, the **GAN-VAE** performed best among the compared models, providing sharper and more **realistic brain tumor images**, with **FID** values between 60 and 100, and improved **SSIM** and **PSNR** scores. Therefore, **hybridization with both adversarial learning and variational encoding** allowed for an optimal balance between **image realism** and achieved smoother reconstructions by incorporating **multi-frequency information**; however, it sometimes resulted in the case of **over-smoothing of textures** and higher values of **FID**, a trade-off for **fine-detail preservation versus stability**. Finally, the baseline **WGAN-GP** showed a relatively higher **FID** and lower **perceptual fidelity**, which again emphasizes its inability to capture the **complex variability within neuro-oncological image data**. Together, these results indicate that the **GAN-VAE framework** provides, so far, the best solution for creating **high-quality synthetic medical images**, marrying **visual realism** with **powerful latent-space representation**. **Synthetic high-quality datasets** of this type are a significant asset in enhancing **Deep Learning pipelines** with **data augmentation**, especially in **rare or sensitive clinical areas**. Future work will focus on extending this study by incorporating **diffusion-based priors** and **spatial attention mechanisms** that further improve **textural fidelity** and **anatomical accuracy**. Overall, the first results of such advanced **hybrid generative models**, such as the **GAN-VAE**, hold great potential to have a **transformational impact on medical imaging** by allowing **data-efficient, privacy-preserving, and diagnostically reliable AI development** for **brain cancer applications**.

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Fig. 1. Pipeline for GAN-VAE.

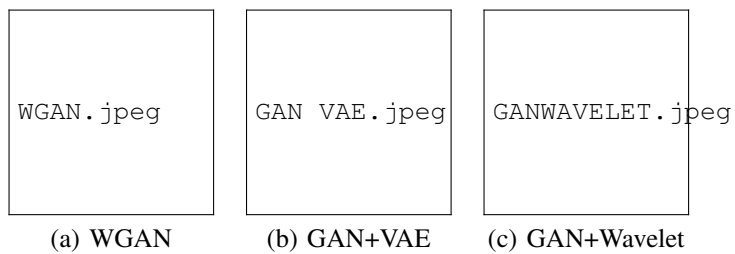


Fig. 2. Qualitative comparison of the generated images from: WGAN (a), GAN+VAE (b), and GAN+Wavelet models (c).